

Section 3

Level 2 Access Requirements

LEVEL 2 ACCESS REQUIREMENTS

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A Level 2 is required to know everything that a Level 1 knows and additional rigging and rescue techniques. We won't rehash all that's been said to this point but assume that if you're reading this you are:

- Moving from Level 1 to Level 2
  - You have enough hours at Level 1 to qualify, 500 minimum for SPRAT, 1000 minimum at IRATA
- Reviewing for renewal of a Level 2 Certificate
- Moving from Level 2 to Level 3 and are refreshing the bits that you've forgotten

In any case, we will start where we started with Level 1 Assessment Requirements.

If you remember, the Level 2 must be able to work with little supervision occasionally. Therefore the Level 2 technician must have a real grasp on keeping others safe by ensuring good rigging, good technique and good rescue ability as well as a bit of managerial skill. Each scheme requires that the Level 2 "demonstrate an understanding of the responsibilities of a Level 2 Technician and how these fit into the overall responsibilities of the employer's Rope Access program".

If we interpret the meaning of that statement we find that the Level 2 should be intimately familiar with how the company operates its' Rope Access program and

further, how everybody (including himself) fits as cogs in this machine. We can see how crucial a good Level 2 is to the organization.

Assessment Requirements

The following Checklist shows what maneuvers, above those required for Level 1 certification, are required for a Level 2 Rope Access Technician to perform during an assessment. The remainder of this section pertains to the specific training that you'll receive to ensure that you can perform these tasks during assessment.

I'm going to stop right here to reinforce a very basic tenement to your successful training: If you're training to become a Level 2 Technician, or you're renewing your certification, it is assumed that ***you are proficient in all Level 1 requirements***. If you are not proficient, delay your training and come in for a few days of refresher training.

You see, if you're not completely proficient in the Level 1 stuff, your training for Level 2 will be very difficult. If you're learning Level 1 stuff over, how much time will be available for the Level 2 stuff you have to learn?

SPRAT changed the Certification Requirements in early 2019. This created a gap that SPRAT Technicians at all levels are going to struggle with for a few years. As a Level 2 candidate in the SPRAT system, you'll appreciate the extra days to get you up to speed with the new Level 1 requirements.

Skill	Req'd
Knots and Hitches	✔
Load Sharing Anchors	✔
Vertical Aid Climbing	✔
Rigging for Hazard Avoidance	✔
Retrievable Rigging	✔
Tensioned Lines (IRATA)	✔
Cross haul	✔
Rescue from Ascent Mode (IRATA)	✔
Rescue Past a Deviation (IRATA)	✔
Rescue Using a Rope to Rope Transfer (IRATA)	✔
Rescue Through a Small Re-anchor (IRATA)	✔
Rescue From Fall Arrest Equipment (IRATA)	✔
Rescue Through Knots (SPRAT)	✔
Rescue From an Aid Climb	✔

Knots and Hitches

You should know plenty of knots from your Level 1 certification. As a Level 2, you'll need to learn about hitches. Friction Hitches are generally a kind of knot used to attach one rope to another in a way that is easily adjusted, but grips when pulled. A prusik knot can be tied to a mechanical advantage system in lieu of a mechanical device to act as the clutch or brake or it can be used as a rope grab for the mechanical advantage system. Another use for a prusik knot is to use it as a foot loop when one is not available to climb the rope.

You'll have to lift somebody up during a rescue, while they're on their ascending gear. This will be a very tough assignment. You may find that carrying a bit of accessory cord tied with a double fisherman's knot will be one of the most beneficial tools on your belt. That bit of cord can be tied onto a rope as a friction hitch (prusik) and help your mechanical advantage system or help you during a tough rescue. Just keep in mind that the prusik should not be used as the sole back-up in a system as the strength of the knot differs with many variables.

Other friction hitches exist that are equally helpful such as an Italian Hitch, commonly called a Munter Hitch, which may be used instead of a belay or descending device if yours broke, for example. An Italian or Munter hitch should be tied off properly to reduce the risk of slippage. A Clove Hitch is an adjustable hitch that is used in various applications but should not be used on life-lines.

A tensionless hitch maintains all of the rope's strength when wrapped around a round object a minimum of 4 times. If you start with a barrel knot, for instance, at the end of the rope and wrap it around the round member a minimum of four times and clip the carabiner through the barrel knot and around the rope, you have a tensionless hitch.

Rigging

Properly rigging ropes for different situations is critical to your safety as a Level 2 technician. There are many options out there, and a bazillion different situations you can come across. Just take your time and consider all of the hazards you're trying to avoid and any that you may be creating.

There are a few things you should always consider when rigging ropes:

1. Always rig systems that have at least two independent anchors.
2. Try to share the load among your anchors. Use double figure of 8 on a bight. They are great for distribution of the load.
3. Consider the entire rope path when evaluating hazards.

4. Consider the swing potential when rigging offset anchor systems.

Re-anchors

We use re-anchors to get around an obstacle or to avoid a hazard. We've all used them, but have we considered the following?

Is there plenty of loop? There is nothing worse than installing a re-anchor without sufficient loop. It makes the passing difficult and the rescue even crazier.

Have you really avoided the hazard? A common use for re-anchors is to install the system so that a technician can access the area under a platform. Keep in mind that there may be a hazard of the rope rubbing on the structure under the platform during the transfer.

Do you have ropes leading to the ground on both sides? It sure is handy when there are ropes on both sides of the system. Install them if convenient.

Deviations

We use deviations to avoid hazards or reposition ourselves. There are a few questions to ask yourself as you plan this rigging:

How great is the angle? We've learned what happens when the angle is too great. Keep the angle equal to or lower than 20°. If the angle needs to be greater, rig a re-anchor instead.

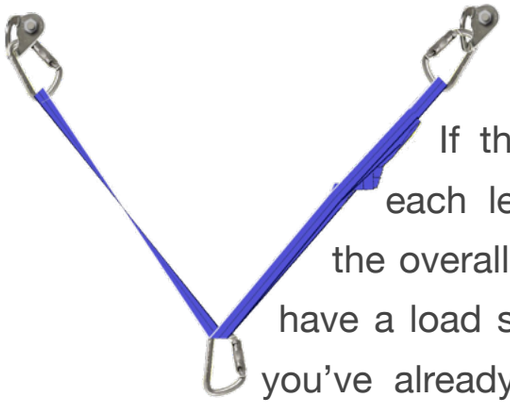
What is the potential that failure of the deviation may cause catastrophic damage to the technician or to equipment? If there is the likelihood of this happening, rig a double deviation.

How ergonomic is the deviation? If the technician will have difficulty passing the deviation, then rig it with longer anchors.

Work Restraint Lines (IRATA)

The intent behind work restraint lines is that we position a worker where a fall hazard exists, but we don't expose the worker to the hazard. Therefore, we need to rig these lines so that the worker is not at risk of falling. You know that the technician will be using a lanyard (usually a cow's tail) to connect to the lines. Try to rig the lines overhead, as far away from the hazard as possible.

Also, make sure the lanyard the technician is using is short enough to access the area, but not long enough to expose the worker to a fall. And finally, you'll have to provide specific work instruction to ensure no more than one worker is attached to a work restraint line, that the use of lanyards is appropriate to the task, and only one person is attached to any segment of the line.



Load Sharing Anchors

If the anchor is designed such that each leg of the anchor system shares the overall load, even if the load shifts, you have a load sharing anchor. An example that you've already seen is the "Y" Hang anchor. Load sharing anchors may include a self-equalizing anchor such as that shown to the left. If the load shifts, the bottom carabiner will shift and change the load on each leg of the anchor.

Plan the use of load equalizing anchors carefully because you could end up making too large an angle on one of the legs. Also, remember that there needs to be redundancy built into your rigged system that will allow for failure.

On "Y" hangs in which the anchors are greater than 2 meters apart, you'll have to take special precautions.

To prevent an uncontrolled swing, tie an alpine butterfly below the "Y" in each line and connect with a carabiner.

Remember: Extend the tail of the 9 at least 3' below the "Y." You'll use these ropes to transfer to the "Y" hang.

Keep your included angles well below 120°.

If you choose to use double figure of 8's on a bite, the angle should not be greater than 60°.

Rope System Pre-Rigged to Lower

Nearly all systems can be pre-rigged to lower. We'll show you several options during class. The intent is to be able to accomplish one or more tasks:

Operate the system to lower a casualty to a position where the casualty is hanging in a vertical presentation with no obstructions below him, or

Lower the system so that the casualty is next to a set of ropes without an obstruction below him, or

Lower the casualty to the ground or other area where emergency medical attention can be administered, or

Transport the casualty to medical attention.

If you remember the hauling system you learned in Level 1 class, you know how to install a basic rope system pre-rigged to lower. The only difference is that you won't attach the casualty to the ropes, you'll have the technician climb your rope system.

There are a few things to modify that may make your life a bit more comfortable:

Use two descenders instead of a descender and a safety back-up. Yeah, it is more difficult to operate, but we're setting this system up for safety, not convenience. Get another person if you need help.

Operate one descender as the lowering system. Put the other descender into belay mode.

Make sure you have enough rope on the back (standing end) side of the system so that you can reach your final destination.

Make sure you have knots at the ends of the standing side of the system.

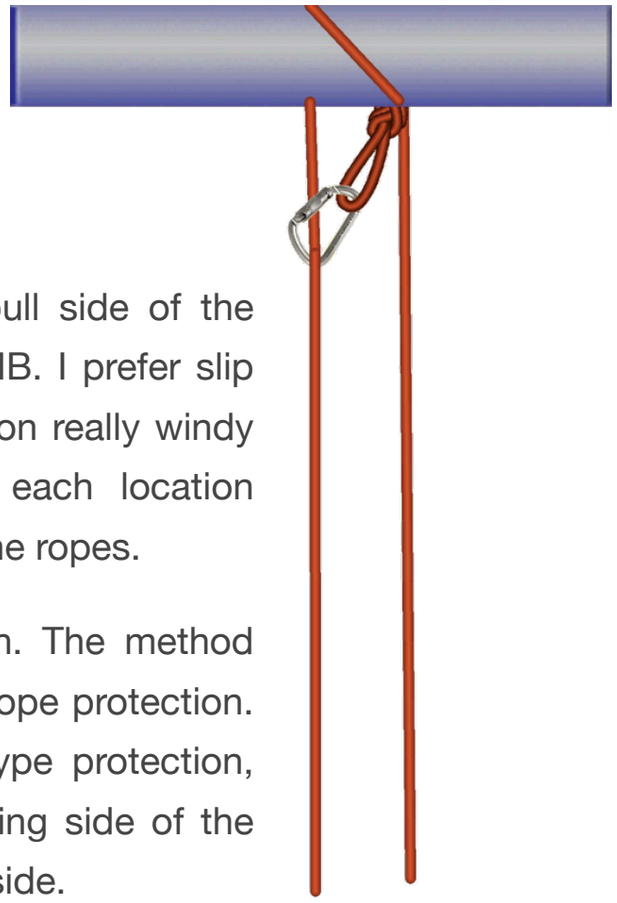
Back-up the descender by tying a slip knot in the standing end of the system (both devices) about 1 meter from the descent device. connect the bight of the slip knot to the anchor.

Retrievable Rigging

Suppose you have to install an anchor and you want to retrieve the ropes from the ground once the work is complete. What do you do? Install a pull through, or retrievable anchor. There are many ways to do this, you'll have to choose the one that works best for the environment you're dealing with. The most basic option is described here:

Tie a Italian Butterfly knot in a rope and throw one end over that section of 6 inch piping. Place a carabiner through the loop of the butterfly and clip it around the other side of the rope. Now pull the rope that is passed through the carabiner until the anchor is in place. You have to do this two times, once for the working line and once for the safety line.

Now's a great time to go over some of the things that will make your retrievable rigging safe and efficient.



1. Tie some sort of knot in the pull side of the ropes to indicate DO NOT CLIMB. I prefer slip knots, but these can come out on really windy occasions. Also, tie knots at each location where a technician may get on the ropes.

2. Use appropriate rope protection. The method described here did not include rope protection. To add simple velcro closure type protection, add the protection to the climbing side of the system, but secure it to the pull side.

3. Do not cross load carabiners, particularly across the gates. This is most important on non-rounded surfaces and small bore piping.

4. Make sure that you can reach both sides of the system when you're on the ground.

5. Tie knots in the ends of the rope, especially if they don't reach the ground.

There are a few alterations you can easily make to this system to help avoid hazards. For instance, if you're concerned about

cross-loaded carabiners or putting a lot of pressure on the butterfly knots, you can add a rope grab (non-aggressive) to each of the climbing ropes, pointed down. Using this method will allow you to position the carabiners and knots away from edges. This method is great when you have to use retrievable rigging through really small anchors such as eye bolts.

One of the best methods for installing a pull through is to add it with protection in place where the climbing ropes are placed below the anchorage and the system is supported by appropriately protected anchors. This should be your default method:

You'll need two ropes, one needs to be more than twice as long as the drop, the other needs to be about ten feet longer than the drop. You'll also need anchor slings that have appropriate protection in place, and they are long enough to fully protect the ropes from the hazard at the anchorage.

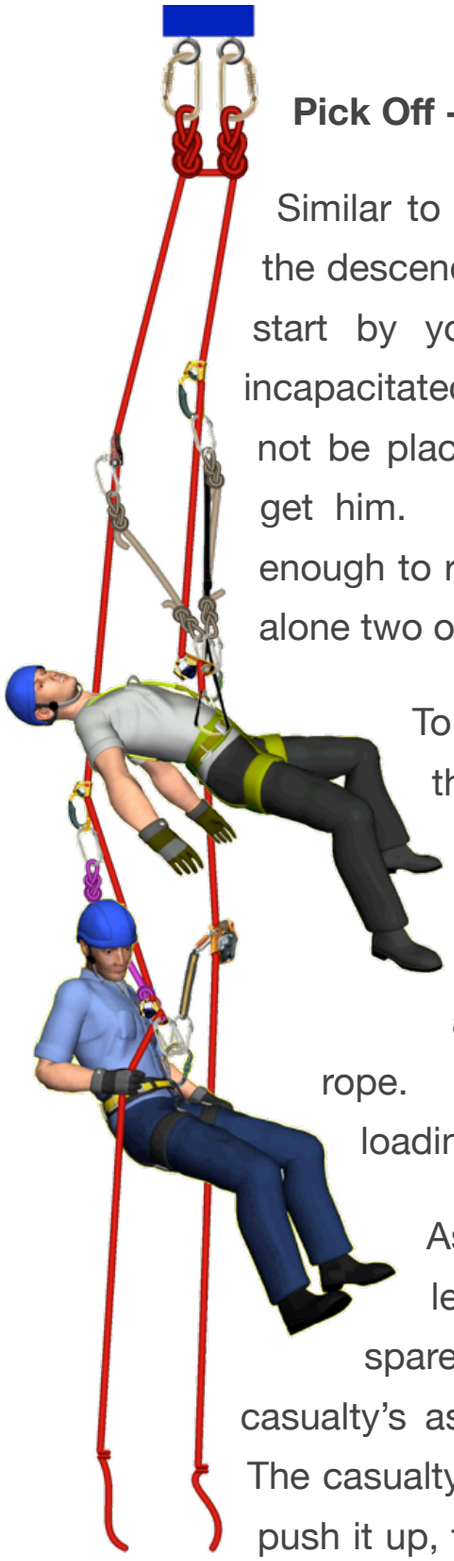
Begin by tying stop knots in the ends of the long rope and then finding its middle. This is a great time to hang the ropes and ensure they reach the ground and there is quite a bit laying on the ground down there. If you're satisfied, pull the ropes back-up, keeping track of where the middle is.

Tie an alpine butterfly about a foot away from the middle on both sides. Connect the shorter rope to the middle rope between the alpine butterflies. This shorter rope will be your pull rope. Use a barrel knot.

So, here's the confusing part: pass the end of your pull rope over the structure and pull until the butterflies are over the anchorage and hanging down. You'll have two ropes hanging on the side of the structure closest to you, and a loop with two butterflies on the other side. Connect an anchor sling to each butterfly using a carabiner. Install a pulley to the climbing side of each of the ropes and connect this pulley to the ends of the anchors. Ensure the anchors are positioned below the anchorage.

If done correctly, you'll be able to rotate the system so that the pulleys are resting on the butterflies under the anchorage. there will be one rope (your short pull rope) that is slack on the top of the anchorage. Protect this rope using appropriate protection if necessary. Tie an appropriate knot in the pull-side of the rope at any location where a technician may get on the ropes, to identify, "Don't climb on this side!"

When you climb on a rope that is attached to a pull through anchor, make sure that you are climbing on the correct side of the anchor. You have to climb on the rope system that will not release the anchors. The side you don't climb should be plainly marked.



Pick Off - Casualty Ascending (IRATA)

Similar to the method used for a pick-off from the descender, pick-off from Chest Ascender will start by you assessing the condition of the incapacitated individual and ensuring that you will not be placing yourself in harm's way if you go get him. That's pretty important as it's hard enough to rescue one person from the ropes, let alone two or more.

To begin, the actual rescue (assuming that you've ensured that you will be safe in doing the rescue), connect your back-up safety device to the casualty's working rope and your ascenders to the casualty's safety rope. This will allow approximately equal loading of both ropes during the ascent.

Ascend to the casualty until you are level with the casualty. Place your spare back-up safety device above the casualty's ascender and push it all the way up. The casualty's hand ascender may be in the way, push it up, too. It's important to keep your back-up device lower than the casualty's hand ascender if you can. in all cases remember that a back-up device should have about 4" of clearance above a toothed device. Remove your first back-up device from the rope.

Ascend past the casualty and change over into your descender. Lock off the descender. Make sure the ropes are not trapped between you and the casualty, move them to one side..

Using one of the casualty's cow's tails, establish a link between the two of you by clipping it into your ventral attachment. This establishes one point of contact to the casualty. Establish the second point of contact by connecting a short link or two carabiners connected to each other from your descender carabiner to the casualty's sternal attachment point.

Make sure you're high enough above the casualty that the short link points down from your descender carabiner to the sternal attachment on the casualty's harness. Doing this will eliminate the need to lower the casualty so much once you take off the chest ascender.

Now it get's fun. The challenge here is that you have to lift the casualty enough to create slack above his chest ascender so that you can remove it.

There are many ways to accomplish this maneuver. We'll just talk about one here, the way we teach this.

Here goes, you'll need a few extra bits of gear; one rope grab, one small pulley and a length of accessory cord, at least 6' long. It might be handy to have a few spare carabiners, too.

You are positioned above the casualty with two points of contact to the casualty. You're also in the descender to make it easy to descend later. Confirm all of this and then remove the casualty's hand ascender and safety back-up. Next, attach a rope grab to the casualty's working line, above his chest ascender. Add the pulley to the rope grab with a carabiner, making sure the pulley is closed, if it has moving side plates, and the carabiner is locked. Using the end of the accessory cord, attach it to the carabiner holding the pulley with an appropriate knot. Run the cord through the casualty's central attachment, through the pulley and repeat twice so that you've created a 4:1 mechanical advantage system.

Pull down on the cord to remove all of the slack and stretch between the casualty's harness and the pulley/rope grab combination. When all of the slack is taken out of the system, and only then, put the hand ascender on to the cord, a bit lower than the pulley. Put your foot into a shortened foot-loop.

Check the position of your back-up device and ensure that it is as high as practicable, but not tight.

Reach down and grab the casualty's harness, preferably at the waist/ventral attachment. While pulling up with your arms, push down with your foot. This action will lift the casualty enough to remove the chest ascender from the rope.

You need to lower the casualty onto the short link. Squeeze the cord at the bottom of the pulley to apply friction to your mechanical system. Take the hand ascender off of the cord while this friction is held tightly. Safely lower the casualty until the system is slack. Take your lifting system off of the line and off of the casualty.

Place an extra carabiner on the carabiner connecting your descender to your harness, on the side of the descender facing you. Feed the slack end of the rope coming from the descender through this carabiner. This will act as extra friction while descending allowing you to control the descent.

It's worthwhile to note here that you must maintain a small fall factor by keeping your back-up device as high as practical. Never allow the back-up device to be low enough to create a fall factor greater than .5.

Unlock your descender, pulling the slack end of the working line upwards so the carabiner you installed applies friction. Descend until you are near the ground maintaining that the casualty does not fully lie on the ground. It is imperative that all of the weight is not taken off of the casualty all at once as, if the casualty has been suspended without movement for a prolonged period of time, the sudden release of pressure could send too much blood

surging through the arteries and cause additional injury. Slowly lower the casualty to the ground all the while moving the legs to restore circulation.

Rescue Through Obstacles

Obstacles can become complex when conducting a rescue, but they all follow essentially the same process. Whether you're rescuing a casualty through knots, deviations or re-anchors, your goal is to transfer from one system to another while the casualty is attached to you.

It is important to point out a few things which will help make things a bit easier and safer:

1. Make sure the short link connection between you and the casualty is under all ropes, particularly loops. You don't want your casualty on one side of the loop and you're on the other. It's not impossible to fix, it's just a pain in the you know what!

2. Always go shopping. Once you've lowered the casualty onto the short link, grab his descender and maybe a back-up device and a few carabiners. You can get them later, but it's easier to get the stuff you need before you really need it.

3. Always use the casualty's cow's tail to make the second connection to you. You'll need your cow's tails to manage your fall factor during transfers.

4. Do not tangle your cow's tails. Look at the path of your cow's tails and ropes. Check often to ensure you haven't tangled something up. I call this "finding hold points". Double check everything every time you put something on a rope or take something off.

5. Always manage your fall factor. Consult the manufacturer's guidance for the back-up you are using.

There are a few rescues that you'll need to do, depending upon if you're training under the SPRAT or IRATA scheme. I'll briefly describe each.

Rope to Rope Transfer with Casualty

This is really a simple technique. There will not be an obstacle in your way. Once you've picked off the casualty, do a bit of shopping. You'll need the casualty's descender.

Add the casualty's descender to your harness. Make sure you capture your ventral D-ring and the short link connection. It's also a good idea that you position the descender on the side of your D-ring in the direction of travel. You don't want to get all tangled up.

Place a safety back-up on the transfer ropes. Pull all of the slack out of the safety line.

Place the other transfer rope into your new descender and remove all of the slack.

Depending upon your position, you may want to climb a bit towards the far anchors. You need to be sure you can transfer safely to the transfer ropes without coming into contact with obstacles or the ground.

Remember that lifting system you installed during the hauling exercises for Level 1? Or maybe you used a lifting system during the pick off from ascent? You'll use it again now.

Place a rope grab above your descender. Take the rope coming from your descender (the standing end) and install a pulley to it. Attach the pulley to the rope grab you just put on. You've just made a lifting system. It looks a lot like the 3:1 that you use in the lifting system, it's not. It's different only because the descender is moving in this example. (Don't call this a 3:1 whilst in Dwayne's presence, lest he gets miffed.)

Place an ascender with foot-loop on the rope you want to pull and go to work. It will be slow going, but not too bad. You can begin to descend using your original descender once you've cleared the obstructions or are sure you won't strike the ground.

Pass a Short Re-anchor with a Casualty (IRATA)

So, a re-anchor is just a series of rope to rope transfers. Your casualty will be one the far side of the re-anchor. You'll have to negotiate the re-anchor to get to the casualty, but you learned how to do that as a Level 1. Just remember to pay attention to the Casualty's equipment. You want to approach the casualty using his safety line as your working line, and so forth.

Pick the casualty off so that he's on your descender carabiner. Do a bit of shopping. This is a great time to make sure your ropes are not under the short link.

You'll want to descend down if you're above the far anchors. Position your descender so that it is slightly lower than the far anchors. Otherwise, don't descend, just carry on.

Add the casualty's descender to your harness. Make sure you capture your ventral D-ring and the short link connection. It's also a good idea that you position the descender on the side of your D-ring in the direction of travel. You don't want to get all tangled up.

Place a safety back-up on the transfer ropes. Pull all of the slack out of the safety line, ensuring that you don't tighten the cow's tail.

Place the other transfer rope into your new descender and remove all of the slack.

You'll install another lifting system like you did in the rope to rope transfer rescue. Climb towards your far anchors. I can't stress this enough: you do not want to descend into the bottom of the loop. Bad things happen there.

Once you've positioned yourself near the far anchors, use your original descender to descend down until the load is on the new descender. Remove it once it is slack. Remove your original back-up as well.

You will be hanging on the loop with exit ropes next to you. Take a look at your ropes. Nothing should be tangled or under your short link.

Remove the original descender and position it on the other side of the tensioned descender, keeping in mind that it has to capture both the central attachment and the short link. Place the empty descender on to the exit ropes, keeping in mind that it should not be placed above the loop ropes. Always move the ropes out of your way. Add your extra safety back-up you took off to the exit ropes.

Complete another transfer and descend to the ground.

Rescue Past Knots (SPRAT)

Your casualty will be above the knots. Ascend past the casualty and perform a descent or ascent rescue, depending upon which equipment the casualty is using. Remember to go shopping.

Descend on your working line until the knot below your descender (on the standing end of the working line) is about 24" from your descender. You'll have to compensate for stretch in the safety line, so stop further away from the knot if there is a large distance between you and the anchorage.

Working with your safety line, move the existing knot so that it is above the height of your descender. I prefer to move the knot by taking up the slack from above the knot and feeding it into the bight of the knot. When done properly, it ensures the damage is isolated and I don't have to leave equipment on a rope later.

Add the casualty's descender to your harness. Make sure you capture your ventral D-ring and the short link connection.

Place a back-up on your working line above your descender. You'll be "making twins" again.

Feed the rope under the knot in your safety line into the descender you just added to your harness. Get as much slack out of the rope as possible.

Transfer the load from the old descender to the new descender by operating the old descender. Remember to use additional friction.

Now that you've transferred from one rope over the knot to the safety line below the knot, take off the slack descender. Remove the original back-up and descend down. Keep in mind, you'll

have a two-person load so it is very important for you to always manage your fall factor, keeping your back-up nice and high.

Negotiate the knot on your safety line by positioning your back-up devices appropriately.

Rescue Past Deviation (IRATA)

So, the deviation is really just like the knots; you'll be transferring from your working line to your safety line. This technique works for single and double deviations.

Once the casualty is hanging on you and you've done a bit of shopping, descend until the top of your descender is a bit lower than the anchor connection of the deviation. You'll have to go a bit lower if the distance to the deviation is longer.

Using the ropes above the deviation, pull yourself into the deviation. You're doing two things here; judging your position to the deviation (you want the descender to be slightly lower than the lowest deviation anchorage), and you're grabbing the ropes below the deviation to lower yourself back out. Adjust your height if necessary (but don't go down too far).

You should be back under your top anchorages for the deviation, with the ropes from below the deviation in your hands, lap, etc. Go ahead and connect them to your harness for now.

Place a back-up above your descender on your working line. Look at that, you've made twins again.

Add the casualty's descender to your harness. Make sure you capture your ventral D-ring and the short link connection. It's also a good idea that you position the descender on the side closest to the deviation. You don't want to get all tangled up.

Grab your safety line (coming from underneath the deviation), and install it into your new descender. Pull all of the slack out of the rope. You'll move towards the deviation. Keep pulling. You want that descender as close as possible to the deviation, about 4" of clearance is good.

Just like you did in the knot rescue, you'll lower your first descender so that the load is transferred to your safety line. You are well protected because you installed a second back-up.

Take off the first descender once it is slack. Keep in mind you have two back-up devices that need attending (you don't want to get hung up on either).

Take off your first back-up and descend down a bit. You want to position yourself so that when you add the safety below the deviation your fall factor will be acceptable.

Add a safety back-up to your safety line below the deviation and remove the safety back-up that is above the deviation. There you go, you've rescued a casualty through a deviation. Good job!

Rescue From Fall Arrest Equipment (IRATA)

Rescuing a casualty who has taken a fall in fall arrest equipment is pretty straight forward. The easiest and quickest way to approach this rescue is by rescue from a separate rope system. Aid climb to the location where the casualty is suspended. Install a rope system and install your descent equipment to it.

Descend to the casualty and make two connections to the casualty as you would any other rescue.

Rig a simple lifting system and lift the casualty, creating slack in their fall arrest lanyard. Unclip the lanyard from the casualty's harness. Lower the casualty onto your short link connection.

Descend to a safe area with the casualty.

Rescue from Aid Climb

The hardest part of rescue from aid seems to be the lifting and lowering part. You see, at best you'll be at the same level as the casualty, probably lower. This creates difficulties because you won't have the same leverage as if you were higher. Another difficulty is that you have to lower the casualty much further because you can't get higher.

Aid out to the casualty's location. Install a set of ropes at the casualty's location, one rope anchored to the anchor the casualty is suspended from, the other on your anchor.

Immediately put the rope located at the casualty into his descender. Remove all of the slack. Place a safety back-up from the casualty's harness on the rope next to you. The casualty will now be connected to the ropes you just hung.

Install a lifting system so that you can lift the casualty, creating slack in his cow's tail. I almost exclusively install a 5:1 mechanical advantage system using accessory cord.

Lift the casualty and release his chest ascender or, oftentimes it's easier to release the carabiner the cow's tail is connected to the anchor with. If you're going to release the carabiner, make sure that the carabiner is either on the front or back, but never in the middle, trapped between two loaded carabiners.

You'll only have to lower the casualty a little bit because you placed the casualty on the ropes you installed. This slick little method will save you lots of heartache if your gear gets caught in pulleys or your accessory cord is short.

Now that the casualty is in descent mode on the ropes, all you have to do is get on the opposite ropes and perform a descent rescue.

You'll most likely need to perform a rope to rope transfer with the casualty to get to the final destination.

Rescue Hauling - Pitch Head Haul

A pitch head haul system is useful for rescuing a casualty quickly to a platform or, with some modification, to another area away from the platform. An advantage of the pitch head haul is that the rescuer is on a platform, most of the time, and that the casualty can be raised or lowered if rigged correctly.

So, let's say that the casualty is immobilized on a set of ropes under a platform, but near the edge. A haul system can be employed that allows the rescuer to bring the casualty to the platform.

Begin by installing a lifting system similar to the one you learned as a Level 1. There are a few differences when installing this system:

1. You won't have to re-direct the ropes through pulleys over the casualty or load. You'll attach your lowering and lifting system over the casualty instead.

2. Rather than attaching the ends of the ropes to the load, you'll use a few rope grabs to attach your system to the casualty's ropes.

3. The rope grab used on the working line is preferred to be a toothed device, such as a hand ascender. This should be positioned all the way down the working line to the casualty's working equipment.

4. The rope grab used on the safety line cannot be aggressive such as a toothed device. This rope grab should be positioned within reach, but low enough to have separation between the rope grab and the back-up safety.

As stated earlier, there are a lot of different ways of doing this and you need to choose the one that makes the most sense in accordance with the circumstances. Many times it will make sense to rig the rescue equipment prior to the access taking place, for instance if there is a chance of needing to rescue somebody in a vertical confined space. In this instance, pre-rigging and testing the system by running a drill may be prudent. Whatever system that you do use, remember to pay attention to loadings. Don't make a bad situation worse by over-stressing a system beyond its safe working load.

Rescue Hauling - Cross Hauling

The Level 2 may be involved in or have to supervise a team hauling rescue. In this case, the roles of each member must be clearly defined and communicated to the team member. Being a supervisor during an emergency is not a pleasant experience under the best of circumstances - even worse if planning isn't involved.

A big part of any Rope Access maneuver is planning. Planning allows you to think ahead and implement steps in order. Team planning is more important because more than one brain is

working on the problem and that's bound to bring up problems because people that don't plan don't think the same.

It's about the expectations that you set for each team member and how those expectations are executed. Pre-planning will help with that. It will also let everyone involved know who the boss is and how to communicate during the task.

A cross haul is simply when you have to get something, the casualty, from one place to another such as across a big gap that exists between the work site and the aid location.

In a cross haul, you'll need to setup two haul systems, one anchored over the load and the other over the intended landing area. Attach the ends of each haul system to the load, using a rigging plate. The rigging plate is very important because you'll be pulling in different directions and the rigging plate is designed for this purpose. Make sure the rigging plate is backed up as well as the rest of your system.

There are a few more details you'll need to think about as you plan this lift and haul:

What is the intended angle you'll achieve at the height of the haul? Remember, the greater the included angle, the more force you'll be putting on each side of the system.

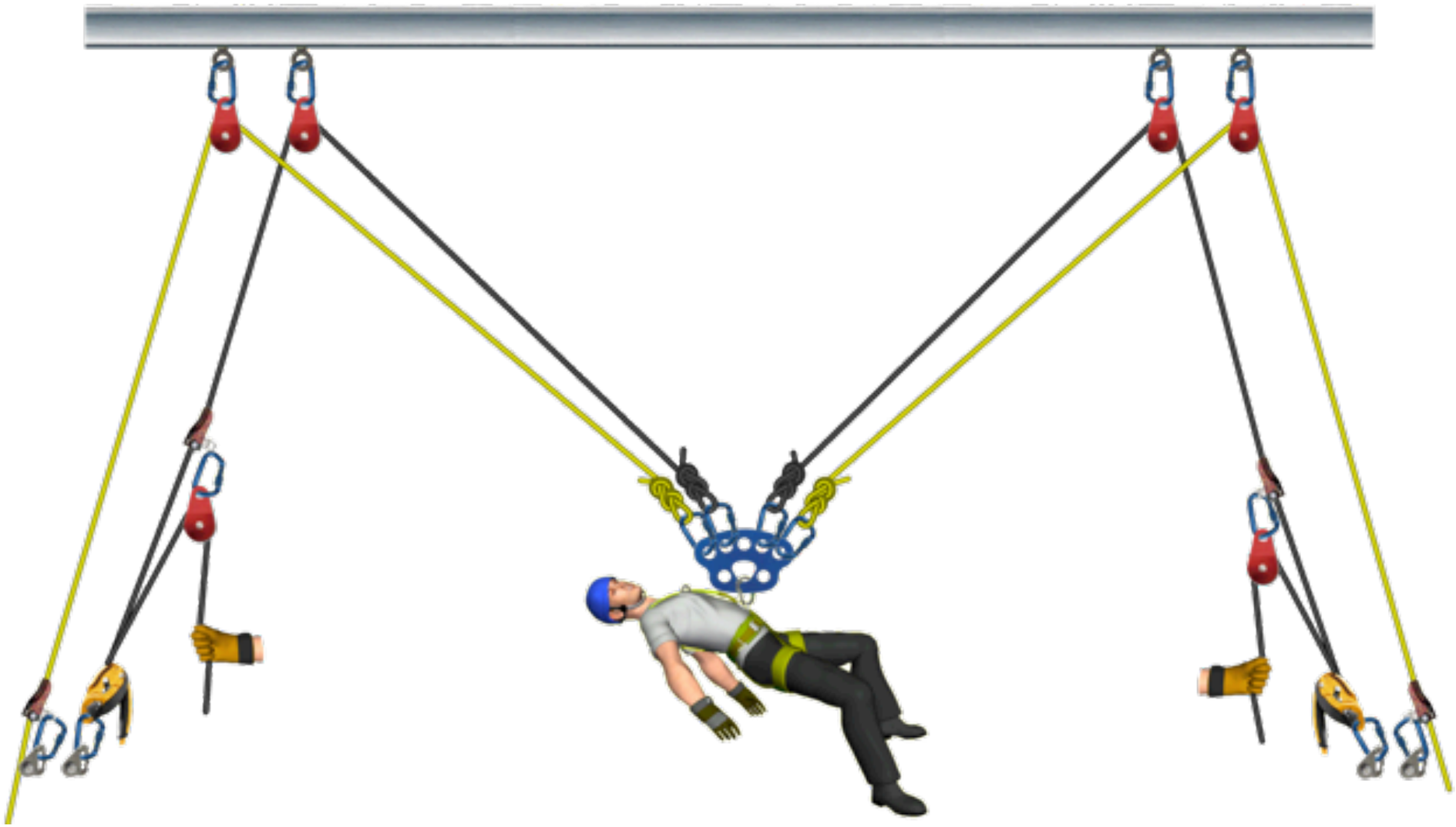
What is the anticipated loading of your system? Will the force be more than the safe working load of your system? In a basic hauling system we use a descender and a safety back-up. It might be more prudent to double up on the descenders if the load is heavy or the angles produce greater forces.

If you are sharing the load among anchors, are you keeping the tensions equal throughout the system? This is particularly important during lowering.

How many people do you need to safely operate the systems?

Operate one side of the system at a time. Lift the load first, move the load to the intended landing area, clearing obstacles as you go, and lower the load using the other system.

Always keep the load as low as necessary to reduce the included angle.



Tensioned Lines (IRATA)

Tensioned lines are used to distribute forces among many anchors and rope systems and to define the load path. That's all they do. Period. Don't make it more complicated than that.

There are a few details that will make your project smoother:

Install a cross haul first. These systems are the motor behind your project. They will be used to lift, transport, and lower your load.

Install a tensioned line at the same level as, or above, your cross haul. You want your tensioned lines to follow the same path as your cross haul.

Note: It is highly recommended that you rig your tension lines as high as practicable.

Keep your load as low as possible. Remember angle loading.

Do not over tension your tensioned lines. A good rule of thumb is one person pulling on a 3:1 mechanical advantage system. Never use more than 2 people pulling on a 3:1 system to tension the lines.

Because the tensioned lines are supporting the forces created, the cross haul should be operated correctly during transport. use mechanical advantage on the haul side if necessary, and put the descender(s) into belay mode on the release side.

Always install at least one control line on the bottom side of a guideline.

Vertical Aid Climb

You'll have to complete a vertical aid climb, which is not any different than a horizontal aid. Remember that you developed a rhythm performing the horizontal aid; L, R, C or R, L, C.

You'll do the same thing on the vertical aid, but the rhythm is Upper, Lower, Center...